Experiment A

```
clear; clc; close all;
rx = sdrrx('Pluto');
rx.CenterFrequency = 2.44e9;
rx.BasebandSampleRate = 5e6;
rx.SamplesPerFrame = 65536;
rx.OutputDataType = 'double';
rx.GainSource = 'Manual';
rx.Gain = 35;
Fs = rx.BasebandSampleRate;
sa = spectrumAnalyzer('SampleRate',Fs, ...
    'SpectrumType','Power density', ...
    'FrequencySpan','Full');
disp('Observe OFF and ON. Stop with Ctrl+C.');
```

```
Observe OFF and ON. Stop with Ctrl+C.
```

```
while true
    x = rx();
    x = x - mean(x); % suppress receiver DC artifact
    sa(x);
end
```

```
## Establishing connection to hardware. This process can take several seconds.
```

```
% After stopping, run: release(rx); release(sa);

% Question 1: 2.4375 - 2.4425 GHz. The peak at 0 Hz represents the DC
% artifact. No observable constant change within the spectrum
```

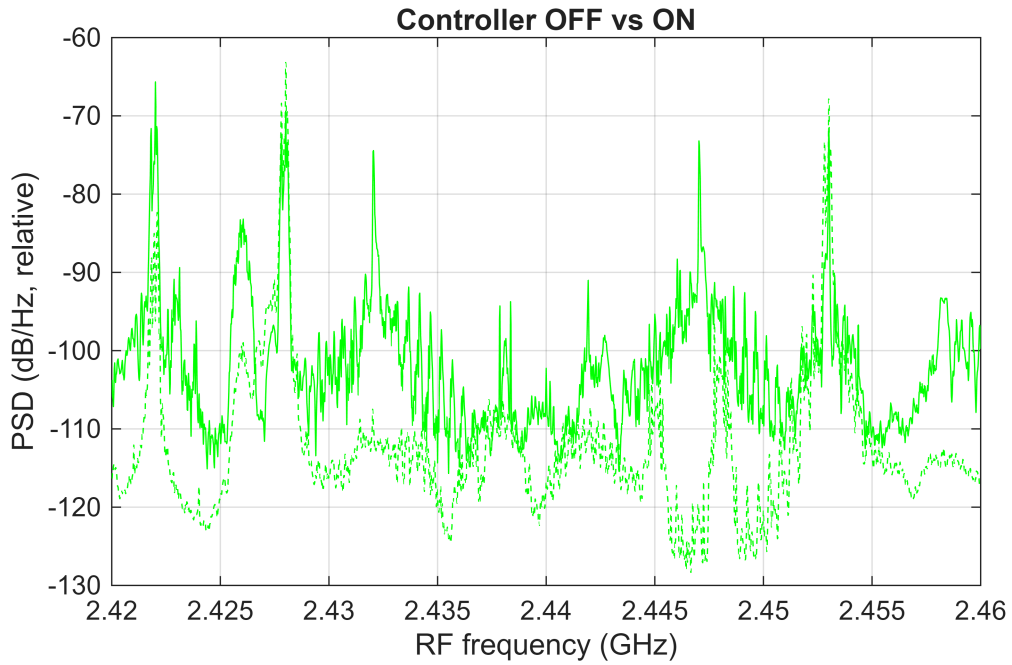
Experiment B

```
clear; clc; close all;
rx = sdrrx('Pluto');
fc = 2.44e9; Fs = 40e6; N = 65536; nFrames = 30;
rx.CenterFrequency = fc;
rx.BasebandSampleRate = Fs;
rx.SamplesPerFrame = N;
rx.OutputDataType = 'double';
rx.GainSource = 'Manual'; rx.Gain = 35;
input('Controller OFF: press Enter to record ');
off = complex(zeros(N*nFrames,1));
for k=1:nFrames
    off((k-1)*N+(1:N)) = rx();
end
```

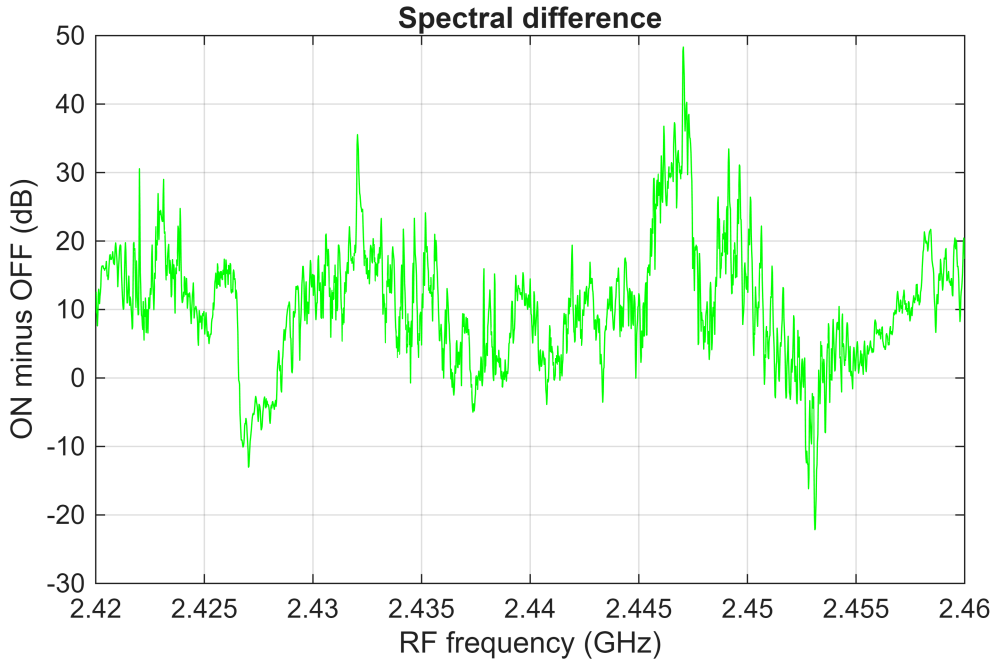
```
## Establishing connection to hardware. This process can take several seconds.
```

```
input('Controller ON, operate steering: press Enter ');
```

```
on = complex(zeros(N*nFrames,1));
for k=1:nFrames
    on((k-1)*N+(1:N)) = rx();
end
release(rx);
off = off-mean(off); on = on-mean(on);
nfft=2048; w=hann(nfft);
[po,f] = pwelch(off,w,nfft/2,nfft,Fs,'centered');
[pn,~] = pwelch(on,w,nfft/2,nfft,Fs,'centered');
figure;
plot((fc+f)/1e9,10*log10(po+eps),'g--'); hold on;
plot((fc+f)/1e9,10*log10(pn+eps),'g-');
xlabel('RF frequency (GHz)'); ylabel('PSD (dB/Hz, relative)');
legend('OFF','ON'); grid on; title('Controller OFF vs ON');
```




```
figure;
plot((fc+f)/1e9,10*log10((pn+eps)./(po+eps)),'g');
xlabel('RF frequency (GHz)'); ylabel('ON minus OFF (dB)');
grid on; title('Spectral difference');
```




```
% Question 1:
```

Experiment C

```
clear; clc; close all;
rx = sdrrx('Pluto');
Fs = 5e6; N = 16384;
rx.BasebandSampleRate = Fs;
rx.SamplesPerFrame = N;
rx.OutputDataType = 'double';
rx.GainSource = 'Manual'; rx.Gain = 35;
freq = 2.402e9:2e6:2.482e9;
nRepeat = 3; framesPerCenter = 12;
Poff = zeros(numel(freq),nRepeat);
Pon = zeros(numel(freq),nRepeat);

for rep=1:nRepeat
    ut(sprintf('Round %d: controller OFF. Enter ',rep));
    for k=1:numel(freq)
        rx.CenterFrequency=freq(k);
        rx(); rx(); % discard initial frames after retuning
        pp=zeros(framesPerCenter,1);
        for j=1:framesPerCenter
            x=rx(); x=x-mean(x);
            pp(j)=mean(abs(x).^2);
        end
        Poff(k,rep)=max(pp);
    end
```

```
    input(sprintf('Round %d: controller ON; operate steering. Enter ',rep));
    for k=1:numel(freq)
    rx.CenterFrequency=freq(k);
    rx(); rx();
    pp=zeros(framesPerCenter,1);
        for j=1:framesPerCenter
        x=rx(); x=x-mean(x);
        pp(j)=mean(abs(x).^2);
        end
    Pon(k,rep)=max(pp);
    end
end
release(rx);

offdB=10*log10(Poff+eps);
ondB=10*log10(Pon+eps);
delta=ondB-offdB;

figure;
plot(freq/1e9,mean(offdB,2),'k--o'); hold on;
plot(freq/1e9,mean(ondB,2),'k-o');
xlabel('Center frequency (GHz)');
ylabel('Peak frame power (dB, relative)');
legend('OFF','ON'); grid on; title('Matched frequency scans');

figure;
imagesc(1:nRepeat,freq/1e9,delta);
axis xy; colormap(gray); colorbar;
xlabel('Scan round'); ylabel('Center frequency (GHz)');
title('ON - OFF difference (dB)');
figure;
plot(freq/1e9,mean(delta,2),'k-o');
xlabel('Center frequency (GHz)');
ylabel('Mean ON - OFF difference (dB)');
grid on; title('Candidate controller frequencies');
[~,ix]=sort(mean(delta,2),'descend');
disp('Top candidate center frequencies (GHz):');
disp(freq(ix(1:min(8,end)))'/1e9);
save('RC_scan_OFF_ON.mat','freq','Poff','Pon','delta','Fs');
```

Experiment D

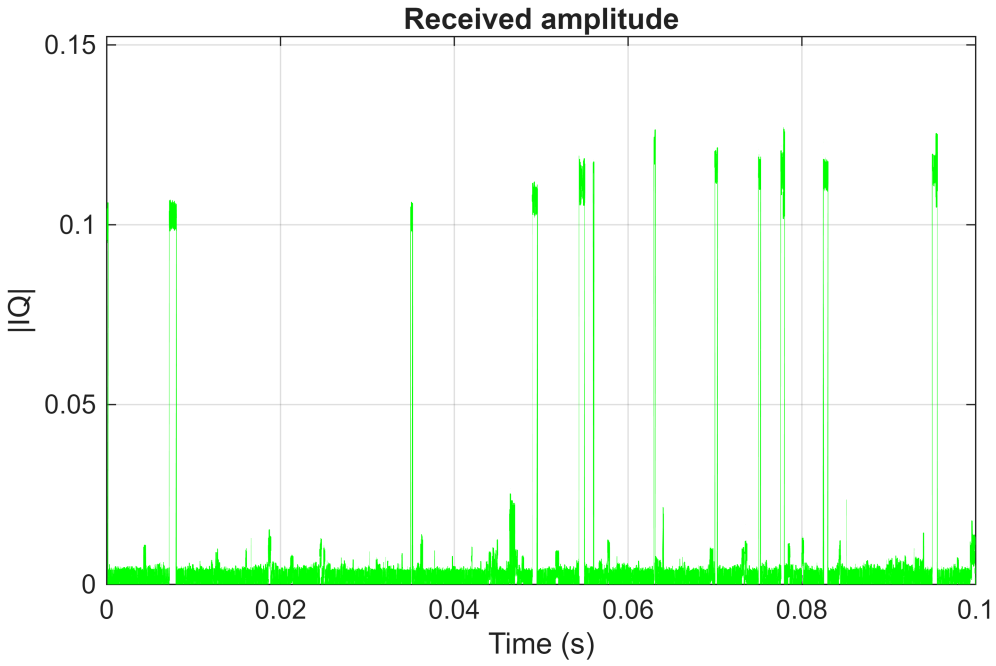
```
clear; clc; close all;
rx = sdrrx('Pluto');
fc = 2.412e9; % replace with your candidate center
Fs = 50e6; N = 50000; nFrames = 100;
rx.CenterFrequency = fc;
rx.BasebandSampleRate = Fs;
```

```
rx.SamplesPerFrame = N;
rx.OutputDataType = 'double';
rx.GainSource = 'Manual'; rx.Gain = 15;
input('Operate controller, then press Enter ');
IQ = complex(zeros(N*nFrames,1));
for k=1:nFrames
    IQ((k-1)*N+(1:N)) = rx();
end
```

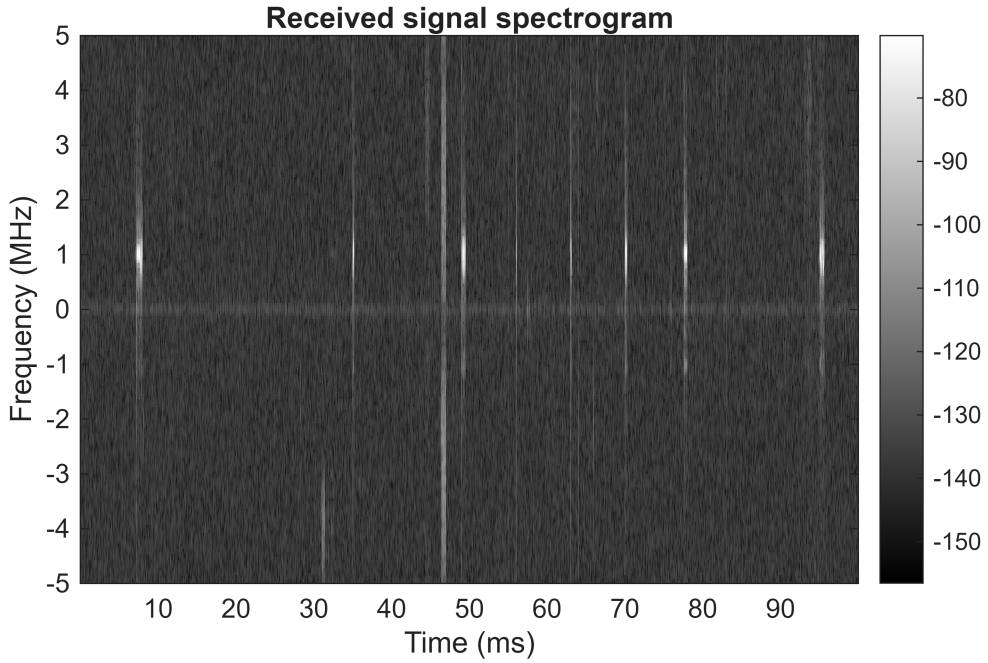
```
## Establishing connection to hardware. This process can take several seconds.
```

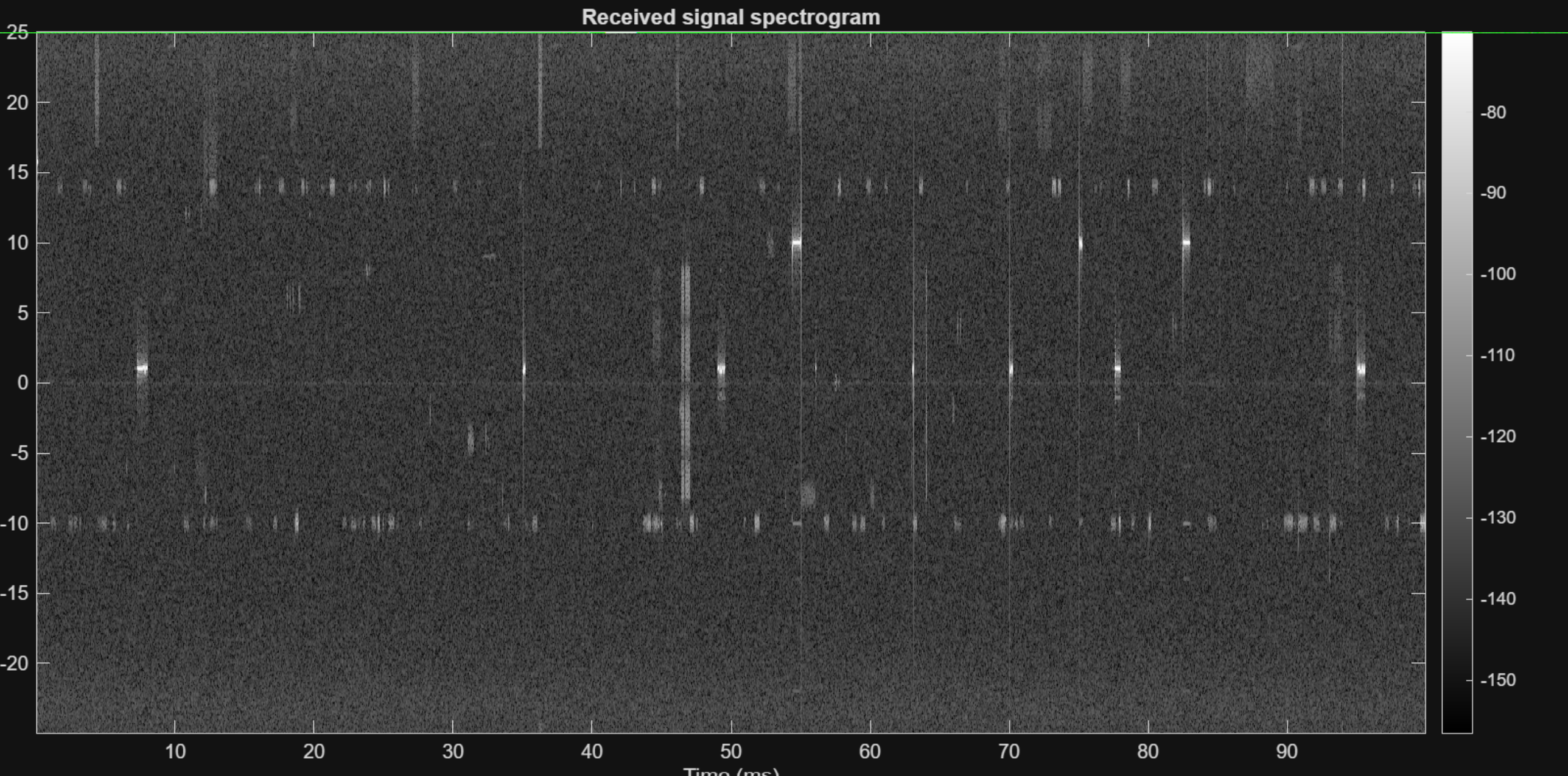
```
release(rx);
IQ = IQ - mean(IQ);
save('RC_IQ.mat','IQ','Fs','fc','-v7.3');

t = (0:numel(IQ)-1)/Fs;
figure;
plot(t,abs(IQ),'g');
xlabel('Time (s)'); ylabel('|IQ|');
title('Received amplitude'); grid on;
ylim([0 1.2*max(abs(IQ))]);
```




```
figure;
spectrogram(IQ,hann(512),384,1024,Fs,'centered','yaxis');
ylim([-5 5]);
title('Received signal spectrogram');
colormap(gray); colorbar;
```


Received signal spectrogram
Frequency (MHz)
5
4
3
2
1
0
-1
-2
-3
-4
-5
10
20
30
40
50
60
70
80
90
Time (ms)
-80
-90
-100
-110
-120
-130
-140
-150


Received signal spectrogram
25
20
15
10
5
0
-5
-10
-15
-20
10
20
30
40
50
60
70
80
90
Time (ms)
-80
-90
-100
-110
-120
-130
-140
-150